```java
import java.util.*;
class Goldbach
{
    static int n;
    Goldbach(int num)
    {
        n=num;
    }

    boolean prime(int x)
    {
        for(int i=2;i<x/2;i++)
        {
            if(x%i==0)
                return false; //checking for prime
        }
        return true;
    }

    void checkGoldbach(int y)
    {
        for(int i=3;i<(y-2);i+=2)
        {
            for(int j=3;j<(y-2);j=j+=2)
            {
                if(((i+j)==y) && (i<=j))
                {
                    if(prime(i)==true && prime(j)==true)
                        System.out.println(i+"+"+j+"="+y); //printing the values of x and y only if both of them are prime
                }
            }
        }
    }

    public static void main()
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("ENTER A NUMBER :");
        n=sc.nextInt(); //accepting a number
        Goldbach ob=new Goldbach(n);
        if(n<9 || n>50 )
        {
            System.out.println("INVALID INPUT. NUMBER IS OUT OF RANGE");
            System.exit(0); //checking if the number is in the range if not the program terminates self
        }

        else
        if (n%2!=0)
```

```java
            System.out.println("INVALID INPUT. NUMBER IS ODD"); //checking if the number is odd
        else
        {
            System.out.print("PRIME PAIRS ARE :");
            ob.checkGoldbach(n);          //calling the checkGoldbach() function if all parameters are met
        }
    }
}


```